

## Project Description

The project is a simple 3D target shooting game developed using OpenGL. The scene will contain a shooting arena where the player can aim and shoot at multiple targets placed at different positions. Targets will move and rotate during the game, while bullets will travel toward the targets. The scene will also include appropriate lighting and different camera views to create an interactive 3D environment.

## Objects

- **Arena:** A rectangular floor surrounded by simple walls to form the main shooting environment.
- **Player Gun:** A small gun constructed from a rectangular body, cylindrical barrel, and a simple handle. The gun can rotate horizontally and vertically while aiming.
- **Targets:** Several circular targets consisting of concentric rings and a central point. Targets will be placed at different distances and heights inside the arena.
- **Target Stands:** Vertical poles with small bases supporting the targets and providing additional structure to the scene.
- **Bullets:** Small spherical or cylindrical objects that move forward from the gun toward the selected target.
- **Arena Objects:** Simple boxes, pillars, and platforms placed around the arena to make the environment visually complete.

## Highlighted Features

- **Target Movement:** Targets will move horizontally or vertically along predefined paths.
- **Object Rotation:** The gun, targets, and other selected objects will rotate to create dynamic movement in the scene.
- **Shooting Animation:** A bullet will appear from the gun and travel toward the target when the player shoots.
- **Hit Effect:** A successful hit will produce a simple visual change in the target, such as scaling, rotation, or temporary disappearance.
- **Lighting:** The scene will include ambient, diffuse, and specular lighting along with directional, point, and spot light sources.
- **Multiple Views:** Front, side, top, and player views will be provided to observe the scene from different perspectives.
- **Camera Movement:** The camera will be able to change position and orientation to provide different views of the arena.
- **Interactive Controls:** Keyboard or mouse controls will allow the player to aim, move the gun, and shoot at targets.